

Tooling. Python 3 / pandas / lingua (language detection) / wordcloud / matplotlib; every number and table body is generated from saved data files, none typed by hand; analysis code and write-up drafted with Claude Code. The code is in the repository https://github.com/intothedeep/mscs_utrgv_cs6330_intro_2_ds, folder hw3/code.

1 Data collection

I compare two music videos: BTS *Dynamite* (video A) and BLACKPINK *Kill This Love* (video B). The data comes from the YouTube Data API through a YouTube Data API v3 key from my own Google Cloud project; the key is not shown. The steps are as follows: Table 1 puts the assignment’s R functions next to the Python scripts that do the same step.

Table 1: Pipeline: the R step of the assignment script and the Python step that replaces it.

Step	R function	Python function	Output file
1. Stats and details	<code>get_stats,</code> <code>get_video_details</code>	<code>p1_collect.fetch_video</code>	<code>data/videos.csv</code>
2. Comments	<code>get_comment_threads,</code> <code>write.csv</code>	<code>p1_collect.fetch_comments</code>	<code>data/comments_<label>.all.csv,</code> <code>data/comments_<label>.csv</code>
3. Terms	<code>Corpus,</code> <code>DocumentTermMatrix,</code> <code>colSums</code>	<code>common.clean_terms,</code> <code>common.count_terms</code>	<code>data/terms_<label>_<version>.csv</code>
4. Word cloud	<code>wordcloud</code>	<code>p2_wordcloud.make_cloud</code>	<code>figures/wordcloud_*.png</code>
5. Comparison	none	<code>p3_compare.py</code>	<code>data/compare_videos.csv,</code> <code>data/compare_terms.csv</code>

1.1 Videos and statistics

Table 2 gives the details and statistics of both videos as of 2026-10-04. The comment count is the total YouTube reports; the script fetched 1,965 top-level comments of video A and 2,127 of video B, and kept 1,000 English comments of each for the English word clouds.

Table 2: Video details and statistics (`data/videos.csv`).

	Video A	Video B
Title	BTS (방탄소년단) 'Dynamite' Official MV	BLACKPINK - 'Kill This Love' M/V
Channel	HYBE LABELS	BLACKPINK
Video ID	gdZLi9oWNZg	2S24-y0Ij3Y
Published	2020-08-21	2019-04-04
Views	2,146,874,108	2,243,728,665
Likes	39,661,294	26,742,200
Comments (YouTube total)	15,898,150	2,500,163
Comments fetched	1,965	2,127
English comments kept	1,000	1,000
As of	2026-10-04	2026-10-04

1.2 Sample

The script reads the top-level comments by relevance order first, then newest first, until 1,000 comments are English (video A) or 1,000 (video B), or a page limit is reached. Each row keeps its order. Table 3 counts the fetched comments by order and detected language.

Table 3: Sample composition by fetch order and detected language (`data/sample_composition.csv`). “Time” is newest first. “No language” means emoji or links only.

Video	Order	Non-English		No language	Total
		English	(language found)		
Video A	relevance	660	413	176	1,249
Video A	time	340	289	87	716
Video A	total	1,000	702	263	1,965
Video B	relevance	579	424	105	1,108
Video B	time	421	434	164	1,019
Video B	total	1,000	858	269	2,127

1.3 First comments

Table 4 and Table 5 show the first five fetched comments of each video, in relevance order. The author handle and the comment ID are dropped, every @mention is replaced by @user, emoji are removed, and long text is cut with “...”. Table 4 and Table 5 stand in for the screenshot of the data.

Table 4: Video A, first five fetched comments (relevance order). “Likes” and “replies” are the comment’s counts; “lang” is the detected language.

#	Order	Published	Likes	Replies	Lang	Comment
1	relevance	2020-08-21	39,987	555	en	I just noticed that I have been listening for 7 hours already Edit: Can’t count anymore lol
2	relevance	2025-09-01	11,459	312	en	To those who are still trying, you’re doing a great job I’m so proud of you. Edit: We did it guys.. congratulations to a...
3	relevance	2025-08-23	8,899	237	en	Just a little bit more. Imagine how happy we will be when it reaches 2B and we are ones of who make it happen.
4	relevance	2026-09-28	26	1	en	BTS just swept at the VMAs, winning SOTY for SWIM, Best K-Pop and Best Group. Congratulations! ARMY is so proud of you.
5	relevance	2020-08-21	20,443	437	tl	Ommg 33 min. 12M WOWWW L’ETS GOOo ARMYYY

Table 5: Video B, first five fetched comments (relevance order). Columns as in Table 4.

#	Order	Published	Likes	Replies	Lang	Comment
1	relevance	2019-06-26	1,058	27	en	*BLACKPINK* - *Kill This Love* M/V 2019-06-26 02am KST:442,647,801 2019-06-27 02am KST:444,465,202 *Today:1,817,401* Bli...
2	relevance	2019-04-04	202	0	en	BP: 'let’s KILL THIS LOVE!' Ikon:’u ’re definitely KILLING ME ’
3	relevance	2026-09-25	215	12	ko	블랙핑크가 많은 악플에 시달리고 있지만, 저는 절대 그들의 노래를 듣는 걸 멈추지 않을 거예요. 여러분, 그들이 한때 K팝을 지배했었다는 걸 잊지 마세요.
4	relevance	2019-04-04	2,164	28	en	Other girlgroups: Love is amazing! Blackpink: Lets kill it!
5	relevance	2021-04-21	453	34	en	Blackpink is the best girl group of K-pop never can change or to be success like them i love you Blackpink

2 Text cleaning and language split

Each comment gets a language label from the `lingua` detector, restricted to 20 candidate languages (`common.LANGUAGES`). With all languages the detector labelled short English comments as Sotho or Tswana, so the candidate list is cut. URLs and @mentions are removed before detection. A comment with no words left gets the empty label and goes to the non-English group.

The word clouds use three versions of the comments of each video:

- **English:** comments detected as English (the first 1,000 of video A, the first 1,000 of video B).
- **Non-English:** every other fetched comment, including those with no language.
- **Mixed:** all fetched comments, English and non-English together.

Table 6 lists the steps from a comment to its terms, as in `common.clean_terms`.

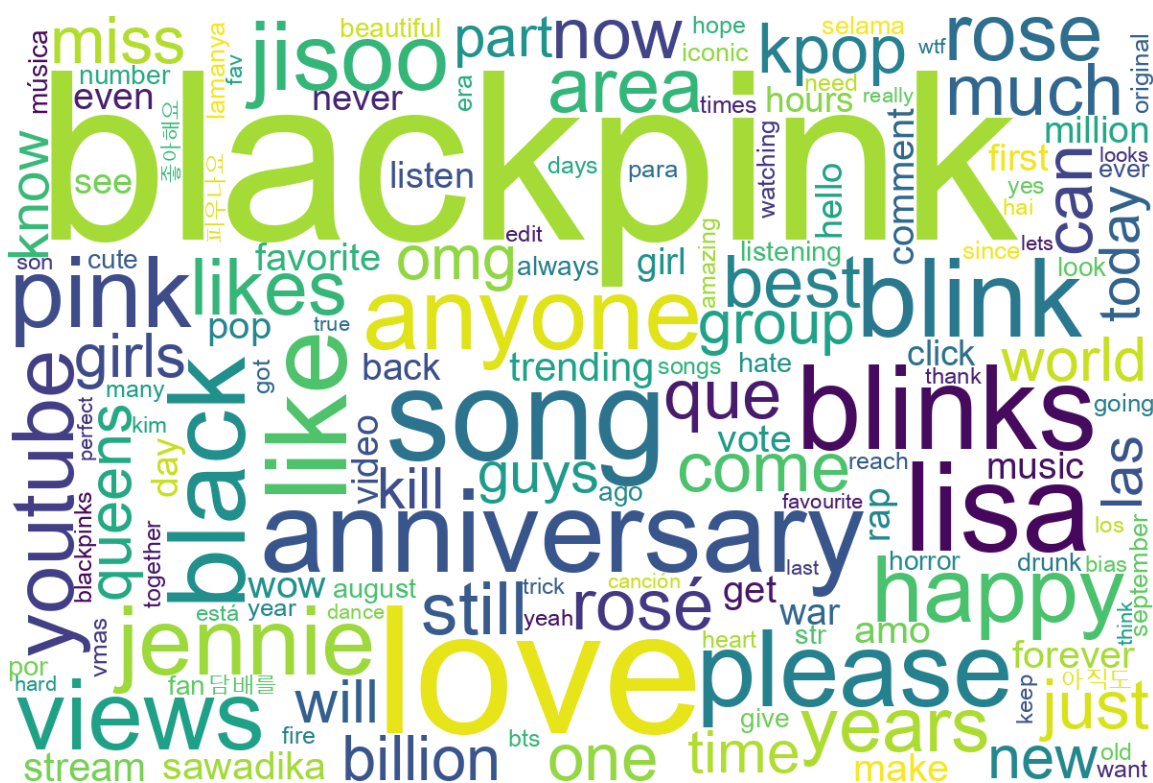


4 Video B word clouds

Figure 4, Figure 5 and Figure 6 show the three clouds of BLACKPINK *Kill This Love*, built the same way as those of video A.



Figure 4: Video B (BLACKPINK *Kill This Love*), English comments ($n = 1,000$).



5 Comparison

Figure 7 puts all six clouds side by side: one row per video, one column per version. Table 7 gives the engagement measures, Table 8, Table 9 and Table 10 the ten most frequent terms of each version, and Table 11 the top 30 terms of each version split into shared and distinctive terms. This section reports results only; the discussion is in Section 6.



Figure 7: All six word clouds. Rows: video A (BTS *Dynamite*) and video B (BLACKPINK *Kill This Love*). Columns: English, non-English and mixed comments.

Table 7: Engagement of the two videos (`data/compare_videos.csv`). “Share fetched” is comments fetched divided by the YouTube comment total; “share English” is English comments kept divided by the same total.

Measure	Video A	Video B
Likes per 1k views	18.47	11.92
Comments per 1k views	7.41	1.11
Share fetched (of YouTube total)	0.0124%	0.0851%
Share English (of YouTube total)	0.0063%	0.0400%

Table 8: Top 10 terms, English comments, with their counts (`data/terms_<label>_en.csv`).

Rank	Term A	Count A	Term B	Count B
1	bts	262	blackpink	351
2	love	148	love	207
3	army	88	song	92
4	song	82	anniversary	83
5	comments	74	blinks	77
6	armys	64	like	76
7	anyone	63	please	72
8	dynamite	61	anyone	66
9	one	58	views	61
10	like	56	black	59

Table 9: Top 10 terms, non-English comments, with their counts (`data/terms_<label>_non_en.csv`).

Rank	Term A	Count A	Term B	Count B
1	que	93	blackpink	97
2	bts	83	lisa	51
3	los	51	que	27
4	amo	38	blink	26
5	army	25	jennie	24
6	dynamite	20	love	22
7	comments	19	jisoo	20
8	jungkook	18	las	19
9	com	18	amo	15
10	como	17	black	14

Table 10: Top 10 terms, mixed comments (all languages), with their counts (`data/terms_<label>_mixed.csv`).

Rank	Term A	Count A	Term B	Count B
1	bts	345	blackpink	448
2	love	161	love	229
3	army	113	song	104
4	que	94	lisa	96
5	comments	93	blinks	85
6	song	92	anniversary	84
7	dynamite	81	like	80
8	armys	80	blink	75
9	anyone	63	black	73
10	like	60	please	72

Table 11: Top 30 terms of each version: in both videos, only in video A, only in video B (`data/compare_terms.csv`).

Version	In both	Only in video A	Only in video B
English	love, song, anyone, one, like, just, views, can, now, still, much	bts, army, comments, armys, dynamite, guys, will, million, today, listening, make, billion, get, jungkook, world, video, comment, watching, people	blackpink, anniversary, blinks, please, black, happy, blink, pink, youtube, lisa, jisoo, years, likes, jennie, area, come, rose, best, miss
non-English	que, los, amo, por, para, música, son, love, está, las	bts, army, dynamite, comments, jungkook, com, como, armys, vamos, todos, esta, soy, con, cute, youtube, kim, estou, então, jimin, porque	blackpink, lisa, blink, jennie, jisoo, black, pink, rose, rosé, sawadika, song, selama, lamanya, hello, 아직도, blinks, omg, wow, 답배를, 피우나요
mixed (all languages)	love, que, song, anyone, like, views, can, now, youtube, still	bts, army, comments, dynamite, armys, one, los, amo, just, jungkook, million, guys, will, jimin, make, today, video, listening, billion, get	blackpink, lisa, blinks, anniversary, blink, black, please, pink, jisoo, jennie, happy, rose, likes, years, area, best, rosé, much, come, kpop

6 Discussion

The discussion is the author’s own comparison and interpretation. It waits for the author’s text; the questions it answers are listed at the end of the report, and it cites Figures 1 to 7 and Tables 7 to 11.

[TODO: Owner: discussion text pending; see the open questions at the end of the report (Section 7).]

7 Limits of the method

- **Top-level comments only.** Replies to comments are not fetched; the replies column in Table 4 is only a count.
- **Sample order.** The sample takes relevance order first, then newest first, so it is not a random sample of all comments (Table 3). The YouTube total is far larger than the sample (Table 2).
- **English stopwords only.** Non-English function words stay in the non-English and mixed clouds (Table 9).

- **No stemming, no merging.** Cleaning keeps accents and does not merge variants, so a name written two ways counts as two terms (for example *rose* and *rosé* in video B).
- **Language detector errors.** One comment of video A is detected as *tl* (Tagalog): “Ommg ... ARMYYY” (Table 4, row 5). It is an English-like fan comment, which shows the detector’s error on short informal text.
- **Cloud rendering.** Arabic words in a cloud are drawn unjoined, because the cloud tool does no text shaping.

Open questions for the author (temporary)

This section is removed before submission.

- B1. [TODO: Owner: The views are close (2,146,874,108 for video A, 2,243,728,665 for video B), but the comment totals differ a lot (15,898,150 for video A, 2,500,163 for video B). What do you think explains this? (Table 2, Table 7)]
- B2. [TODO: Owner: Video A ranks army, comments, million, views. Do the comments talk about the song, or about the fandom and its counts? (Table 8, Table 11)]
- B3. [TODO: Owner: Video B ranks anniversary and happy. What anniversary do you think the comments mean? (Table 11)]
- B4. [TODO: Owner: love ranks second in both. Is it the same meaning in both? (Video B’s title holds “Love”.) (Table 8)]
- B5. [TODO: Owner: Video B names four members, video A mostly one. What does that say, if anything? (Table 11)]
- B6. [TODO: Owner: Which of the three cloud versions best supports your comparison, and why? (Figures 1 to 7)]
- B7. [TODO: Owner: What does the non-English cloud say about each audience? Is it useful given the stopwords limit? (Figure 2, Figure 5, Table 9)]
- B8. [TODO: Owner: Does mixing relevance order and newest-first order change what the clouds show? (Table 3)]
- B9. [TODO: Owner: What is your overall conclusion: how are the two comment sections alike, and how do they differ?]